

Introduction to RMarkdown - PDF/Word

Dr. Mutua Kilai, PhD

September-December 2026

Department of Pure and Applied Sciences

Creating PDF Document

- To create a PDF document from R Markdown you specify the `pdf_document` output format in the YAML metadata

```
1 ---
2 title: "SPS 2444"
3 author: "Student Name"
4 date: "September 04, 2024"
5 output: pdf_document
6 ---
```

- Within R Markdown documents that generate PDF output, you can use raw LaTeX, and even define LaTeX macros.

Note

Note that PDF output requires installation of LaTeX

Table of Contents

- You can add a table of contents using the `toc` option and specify the depth of headers that it applies to using the `toc_depth` option.

```
1 ---
2 title: "SPS 2444"
3 author: "Student Name"
4 date: "September 04, 2024"
5 output:
6   pdf_document:
7     toc: true
8     toc_depth: 2
9 ---
```

Number sections

- You can add section numbering to headers using the `number_section` option

```
1 ---
2 title: "SPS 2444"
3 author: "Student Name"
4 date: "September 04, 2024"
5 output:
6   pdf_document:
7     toc: true
8     toc_depth: 2
9     number_sections: true
10 ---
```

Figure options

- `fig_width` and `fig_height` can be used to control the default figure width and height
- `fig_crop` controls whether the `pdfcrop` utility if available in your system is automatically applied to PDF figures.
- `fig_caption` controls whether figures are rendered with captions

```
1 ---
2 title: "SPS 2444"
3 author: "Student Name"
4 date: "September 04, 2024"
5 output:
6   pdf_document:
7     fig_width: 7
8     fig_height: 6
9     fig_caption: true
10 ---
```

Data Frame printing

- You can enhance the default display of data frames via the `df_print` option
- The various options available include:
 - i. default - call the `print.data.frame` generic method
 - ii. Use the `knitr::kable()` function
 - iii. Use the `tibble::print.tbl_df()`

Word Document

- To create a Word document from R Markdown, you specify the word_document output format in the YAML metadata of your document:

```
1 ---  
2 title: "SPS 2444"  
3 author: "Student Name"  
4 date: September 22, 2024  
5 output: word_document  
6 ---
```

- The most notable feature of Word documents is the Word template, which is also known as the “style reference document.”
- You can specify a document to be used as a style reference in producing a *.docx file (a Word document).